

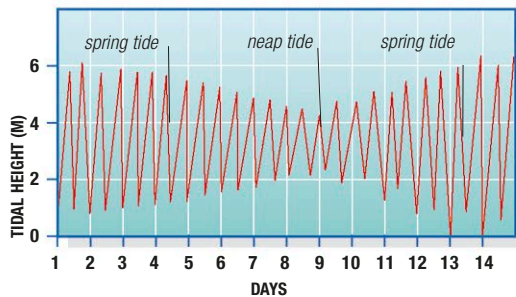
Life on coasts and coral reefs

Compared with the open sea, coasts and coral reefs are disproportionately rich in marine wildlife. Rocky coasts and mudflats abound with seabirds and invertebrates, while coral reefs probably contain at least a third of the world's fish species. Conditions in coasts and reefs are variable, so the animals that use them are often extremely specialized. Some roam over large areas of the shore, but for most, a distance of just a few yards makes the difference between an ideal habitat and one in which it is impossible to survive.

Adapting to tides

For coastal animals, life is governed by the rhythm of the tides. As well as adapting to the twice-daily ebb and flow, they have to adapt to the drawn-out rhythm of extra-large spring tides, which are normally 14 days apart. Knowing where the tide stands is important because animals caught unprepared run the risk of either drying out or drowning.

Some shore animals, such as barnacles, adjust their behavior according to whether or not they are submerged. Most aquatic animals, however, are much more sophisticated, reacting to their internal biological clocks,



TIDAL RHYTHMS

When the sun and moon line up with the earth, extra high and low "spring" tides occur. When they are at right angles, the result is a neap tide, with minimal variation.

MANGROVE SWAMPS

Mangrove swamps are formed by trees that are adapted for life in salty intertidal mud. Found only in the tropics and subtropics, they play an important ecological role by stabilizing the coast and providing inshore nurseries for fish and other marine animals. Most mangroves develop arching prop roots, which are exposed when the tide is out and submerged when it is in. These provide anchorage for mollusks, and act as convenient perches for mudskippers—the air-breathing fish that thrive in this habitat. Mangrove foliage is tough and leathery, and eaten mainly by insects; but the dense canopy attracts large numbers of birds, which use mangroves as breeding sites and overnight roosts. The mud in mangrove swamps is deep and often foul smelling, but it is rich in organic matter that is replenished each day by the tide.

Until the late 20th century, mangroves escaped much of the large-scale deforestation in the tropics. But, with the increase in shrimp farming, large areas of mangroves have been cut down, raising concerns about the long-term impact on coastal wildlife.



CLOSED ANEMONE

OPEN ANEMONE

PREVENTING WATER LOSS

To protect themselves from drying out at low tide, many sea anemones withdraw their tentacles, turning into jellylike blobs that can be seen on coastal rocks. They often live under overhangs, which keeps them out of direct sunlight.

which march in step with the tides. Even if they are taken away from the shore and placed in a tank, their tidal clocks continue to tick.

This built-in ability to keep time enables animals to anticipate events. For example, submerged limpets crawl over rocks to graze on microscopic algae. But since they are vulnerable away from their niche on the rock, they have to return home before the sea ebbs away. Their biological clocks tell them when to head back, ensuring that they are securely in place, and airtight, before the tide goes out. Conversely, the fiddler crab's biological clock prompts it to come out and feed at low tide and then return to its underground burrow before the incoming tide engulfs it.



SWAMP VEGETATION

Mangroves are the only trees that can grow in mud periodically flooded by the tide. Few animals eat the trees' leaves, partly because they are often rich in salt, but these coastal forests are important breeding grounds for birds and marine animals.

Feeding on the shore

Coasts provide a much greater variety of food than the open sea. High up the shore, an abundance of animal and plant remains left in strandlines by the retreating tide are consumed by beach-hoppers, springtails, and other scavengers, which are then preyed upon by gulls and waders.

In the intertidal zone, filter feeders are common. Unlike the giant filter feeders of the open sea (see p.74), coastal ones are generally small and often spend their entire adult lives fixed in one place. Mussels and other bivalves, which filter particles of food using modified gills, are examples of these. Barnacles have a different filtering technique: despite their resemblance to mollusks, barnacles are crustaceans, with a set of feathery legs. At high tide, the legs protrude from the barnacle's case to collect food particles.

Rocky shores are home to some fast-moving swimmers as well as to many other animals that take a more leisurely approach to finding



FEEDING AT THE SHORE

Sandpipers wade into the sea in their search for food. Like many coastal waders, they follow each wave as it retreats, and then they scuttle back up the shore as the next one breaks.

food. Starfish and sea urchins are among the slowest, crawling over rocks on hundreds of fluid-filled feet.

Living in mud and sand

Compared with rocky shores, coastal mud and sand seem to harbor only a limited amount of animal life. But appearances are deceptive. Mud in particular often teems with hidden life feeding on organic matter brought in by the tide.

One of the advantages of living below ground is that despite the attacks of curlews and other long-billed birds, it provides good protection from predators. The chief disadvantage is that the surface is constantly shifting, cutting off buried animals from the water above—and from oxygen and food.

Some of these buried animals have specialized body parts that enable them to connect with the surface. Clams, for example, have leathery tubes or siphons. In many species, the siphons can be retracted inside the shell, but in some, they are too long to be stowed away. Species that do not have such accessories often live in burrows. The lugworm